

Clustering

Advanced Statistics Lab

Claas Heuer

Clustering

Clustering or *Unsupervised Learning* deals with learning grouping structures from *unlabelled* data.

Unlabelled data means the absence of trait or phenotypic observations in a breeding context while having *features* available on a set of individuals.

Features And Labels

Illustration on a dataset that contains *features* and *labels*.

The features here are

Sepal.Length, Sepal.Width, Petal.Length and
Petal.Width.

The labels for those feature observations are the **Species**.

Labelled Data

```
library(pacman)
p_load(ggplot2, knitr, plotly, lubridate, tidyverse, rstan)

# load dataset
data(iris)

# check out features and labels
glimpse(iris)
```

Unlabelled Data

Now we treat the features as coming from unlabelled individuals, meaning we don't have information on the grouping factor (Species)

```
# load dataset
data(iris)

# check out features and labels
glimpse(iris %>% select(-Species))
```

Univariate Clustering

An initial approach is to investigate the individual features in isolation from each other and try to infer a grouping structure.

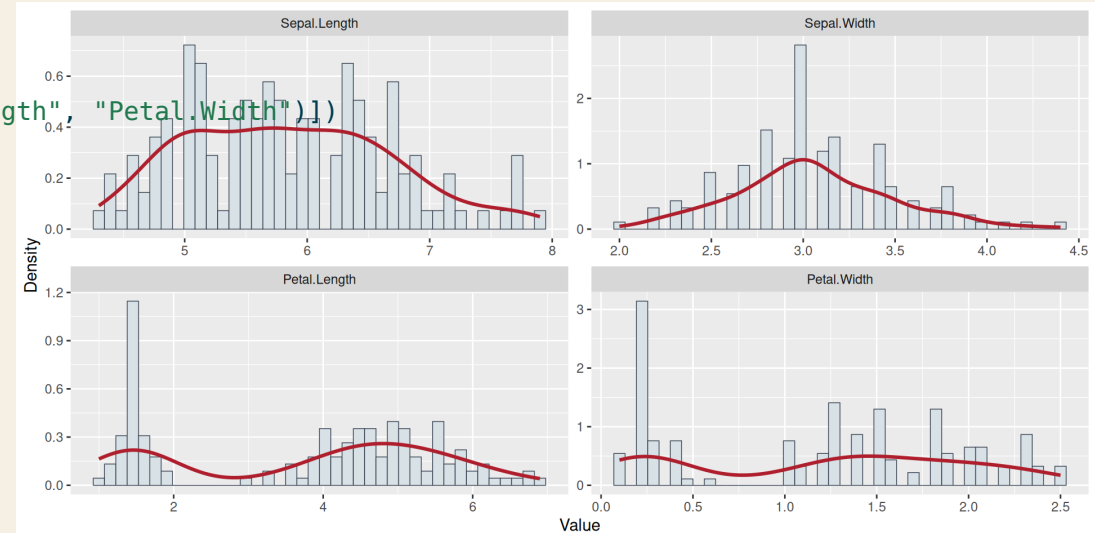
```
dplot = pivot_longer(iris, -Species, names_to = "Feature", values_to = "Value")
p = ggplot(dplot, aes(x = Value)) +
  geom_histogram(aes(y = ..density.., fill = Feature), bins = 40) +
  geom_density(color = "#b3212d", linewidth = 1.1) +
  facet_wrap(~ Feature)
p
```

Feature Distributions

```
library(ggplot2)
data(iris)

dplot = stack(iris[, c("Sepal.Length", "Sepal.Width", "Petal.Length", "Petal.Width")])
colnames(dplot) = c("Value", "Feature")

ggplot(dplot, aes(x = Value)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 40,
    fill = "#d9e2e7",
    color = "#374151",
    linewidth = 0.25
  ) +
  geom_density(color = "#b3212d", linewidth = 1.1) +
  facet_wrap(~ Feature, scales = "free") +
  labs(x = "Value", y = "Density")
```



Normal Mixture Model

Some features indicate that the data had been generated not from a single normal process, but potentially a series of processes.

$$p(\mathbf{x}) = \sum_{i=1}^K \lambda_i f_i(\mathbf{x}), \quad \text{with } \sum \lambda = 1.$$

Univariate normal distribution: $f(x) \sim N(\mu, \sigma^2)$.

Simulate Mixture Data

```
library(ggplot2)
theme_set(theme_minimal(base_size = 16))

# some random data
set.seed(100)
N <- 1000

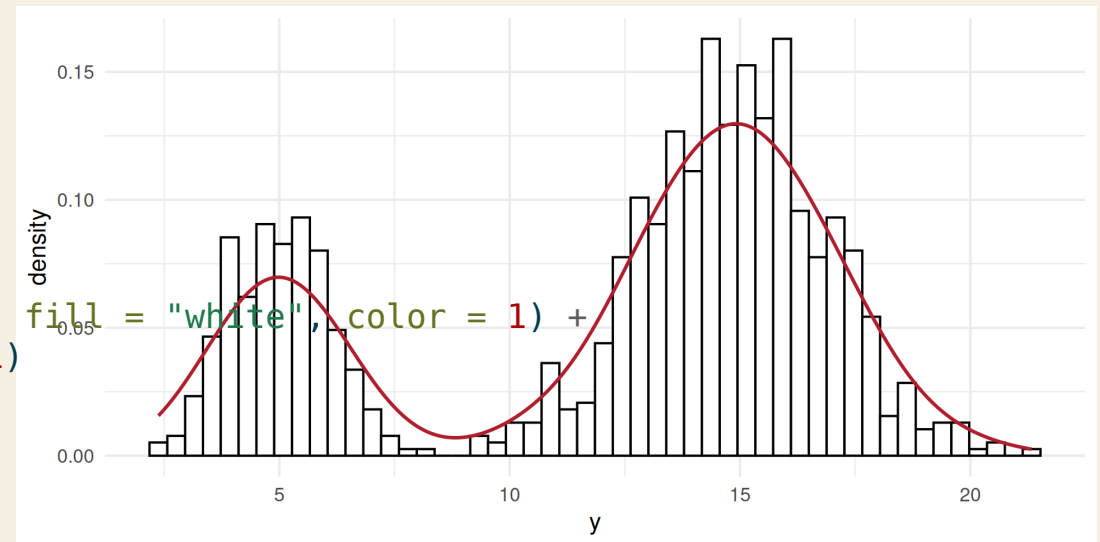
# mixture probability for the first component
theta <- 0.3
mu1 <- 5
mu2 <- 15
sd1 <- 1
sd2 <- 2
y <- ifelse(runif(N) < theta, rnorm(N, mu1, sd1), rnorm(N, mu2, sd2))

dat = list(
  N = length(y),
  y = y,
  k = 2
)
```

Raw Mixture

```
dplot = as.data.frame(dat)

ggplot(dplot, aes(x = y)) +
  geom_histogram(aes(y = ..density..), bins = 50,
  geom_density(color = "#b3212d", linewidth = 1.1)
```



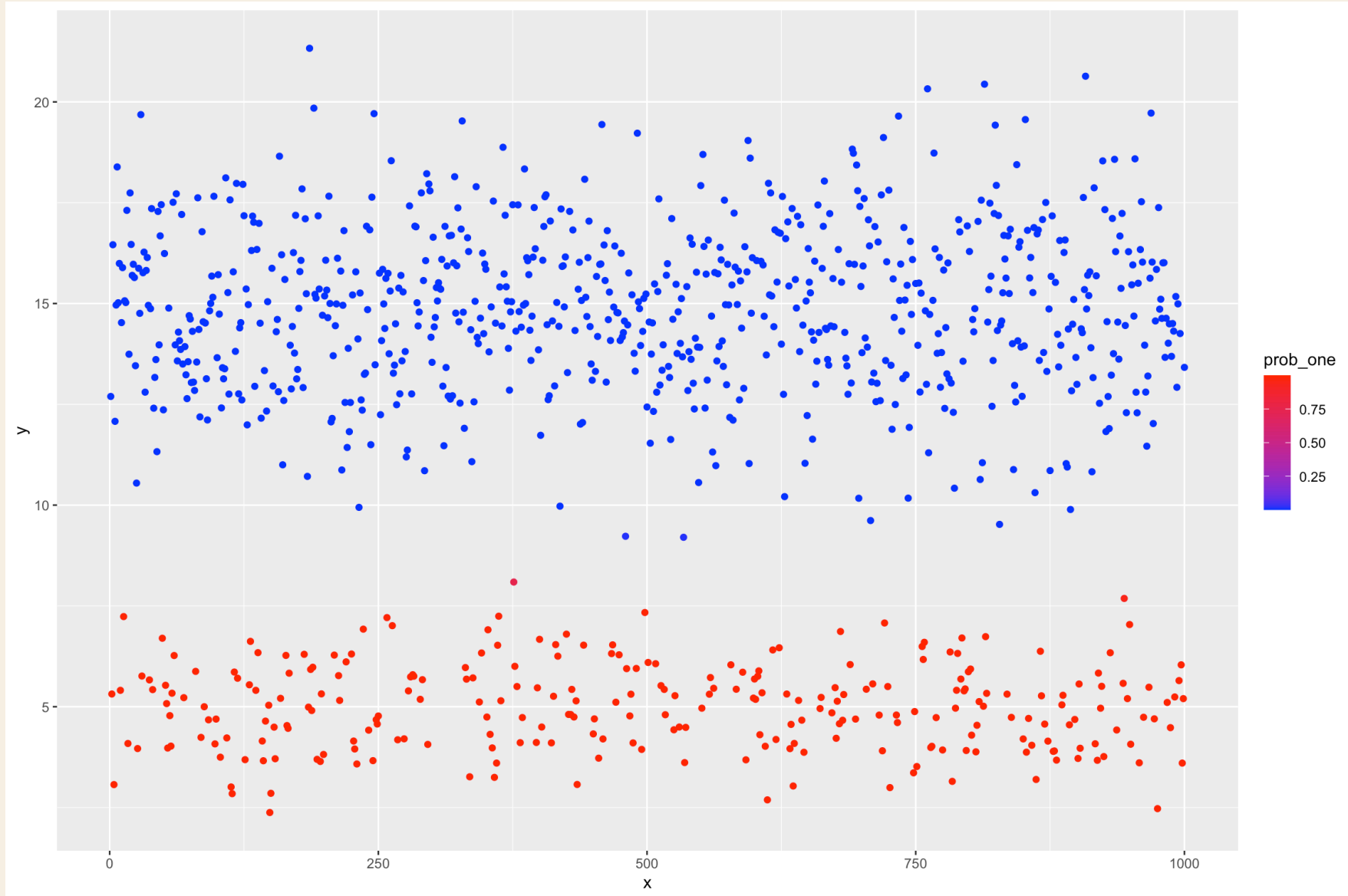
Stan Mixture Program

```
hcode =  
"  
data {  
  int<lower=1> N;  
  int<lower=1> k;  
  real y[N];  
}  
parameters {  
  simplex[k] theta;  
  real mu[k];  
  real<lower=0> tau[k];  
}  
model {  
  real ps[k];  
  for (i in 1:k){  
    mu[i] ~ normal(0, 1.0e+2);  
  }  
  for(i in 1:N){  
    for(j in 1:k){  
      ps[j] <- log(theta[j]) + normal_log(y[i], mu[j], tau[j]);  
    }  
    increment_log_prob(log_sum_exp(ps));  
  }  
  tau ~ cauchy(0,5);  
}  
"
```

Fit Mixture

```
hlm = stan(  
  model_name = "Gauss mixture",  
  model_code = hcode,  
  data = dat,  
  iter = 2000,  
  warmup = 1500,  
  verbose = FALSE,  
  chains = 1  
)  
  
out <- rstan::extract(hlm)  
  
thetas = colMeans(out$theta)  
mus = colMeans(out$mu)  
sds = colMeans(out$tau)
```

Fitted Mixture Probabilities



Fitted Mixture Densities

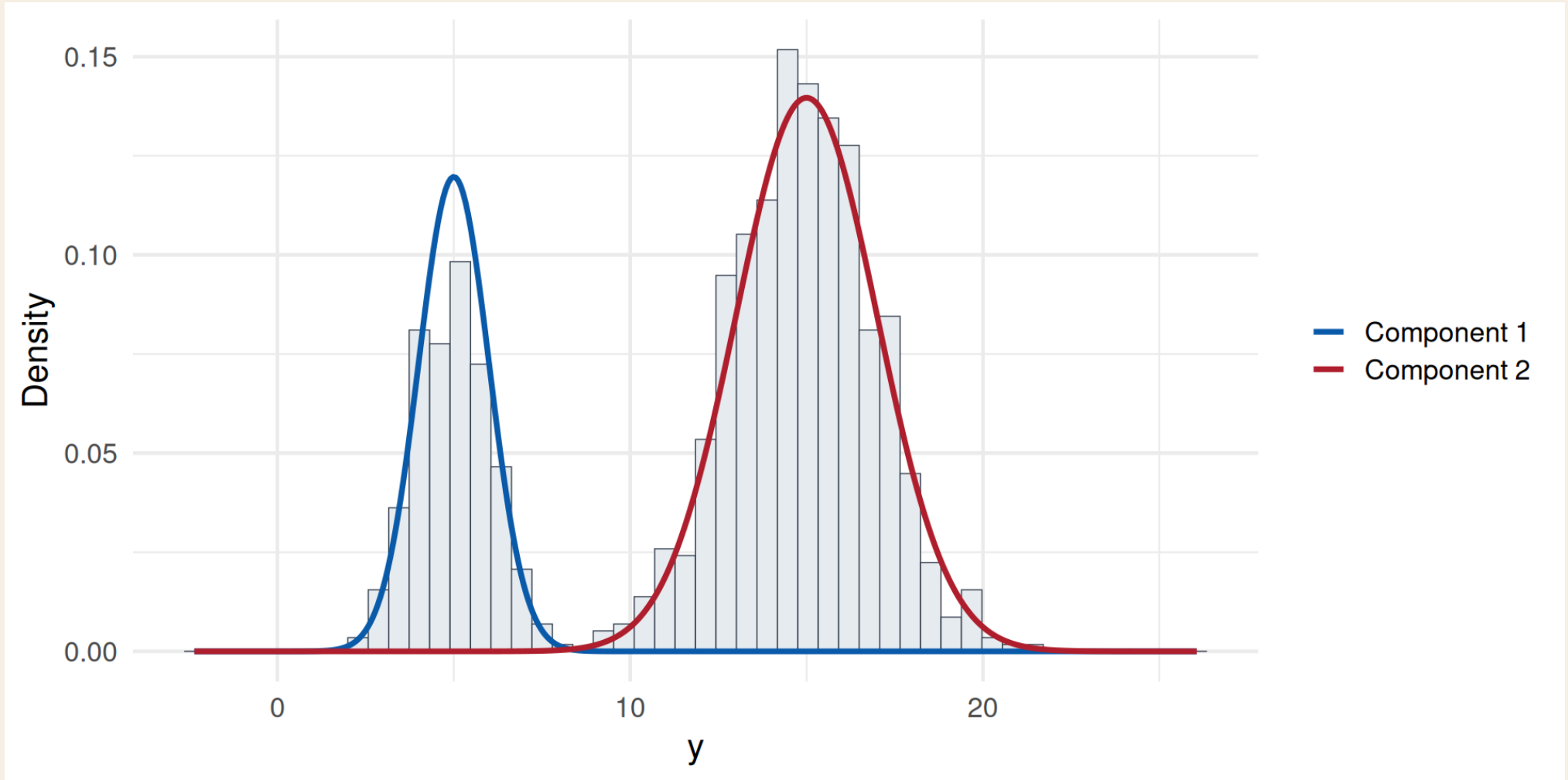
```
sim_density = rbind(
  data.frame(
    component = "Component 1",
    y = seq(min(y) - sd(y), max(y) + sd(y), length.out = 400),
    lambda = theta,
    mu = mu1,
    sigma = sd1
  ),
  data.frame(
    component = "Component 2",
    y = seq(min(y) - sd(y), max(y) + sd(y), length.out = 400),
    lambda = 1 - theta,
    mu = mu2,
    sigma = sd2
  )
)

sim_density$density = with(sim_density, lambda * dnorm(y, mu, sigma))

mixture_density_plot = ggplot(dplot, aes(x = y)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 50,
    fill = "#e8edf0",
    color = "#374151",
    linewidth = 0.25
  ) +
  geom_line(
    data = sim_density,
    aes(x = y, y = density, color = component),
    linewidth = 1.25,
    inherit.aes = FALSE
  ) +
  scale_color_manual(values = c("Component 1" = "#0b5cad", "Component 2" = "#b3212d")) +
  labs(x = "y", y = "Density", color = NULL)

mixture_density_plot
```

Fitted Mixture Densities Plot



Mixture Probability Helper

```
GetProbNormalMixture <- function(y, lambda, mu, sigma) {  
  
  lik <- array(as.numeric(NA), dim = c(length(y), length(lambda)))  
  for(i in 1:length(lambda)) lik[,i] <- lambda[i] * dnorm(y,mu[i],sigma[i])  
  probs <- lik / rowSums(lik)  
  return(probs)  
  
}  
  
dnormLambda <- function(x, lambda, mu , sigma) {  
  return(lambda * dnorm(x, mu, sigma))  
}
```


Apply To Iris

First, bring everything together into a function that we can call conveniently on different data and different number of mixture components.

```
RunMixture = function(y, k = 2, niter = 2000, warmup = 1500, seed = NULL) {  
  # stan program, fit, posterior extraction, plots  
}  
  
seed = 242151  
  
sl_mixture = RunMixture(iris$Sepal.Length, k = 2, seed = seed)  
sw_mixture = RunMixture(iris$Sepal.Width, k = 2, seed = seed)  
pl_mixture = RunMixture(iris$Petal.Length, k = 2, seed = seed)  
pw_mixture = RunMixture(iris$Petal.Width, k = 2, seed = seed)
```

Iris Mixture Densities

```
iris_features = iris[, c("Sepal.Length", "Sepal.Width", "Petal.Length", "Petal.Width")]

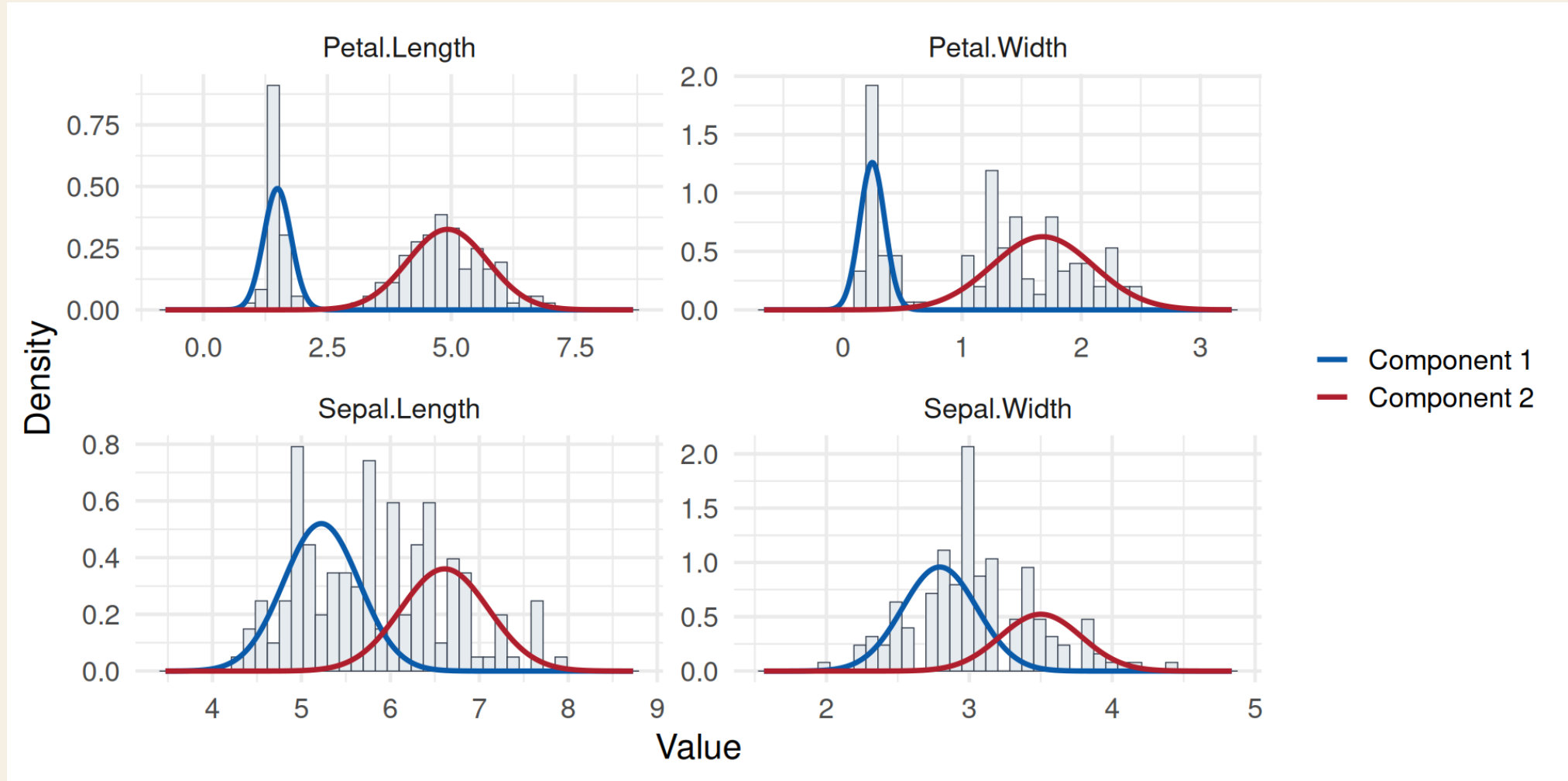
iris_density = do.call(
  rbind,
  lapply(names(iris_features), function(feature) {
    out = mixture_preview(iris_features[[feature]], k = 2, seed = 242151)
    out$Feature = feature
    out
  })
)

iris_long = stack(iris_features)
colnames(iris_long) = c("Value", "Feature")

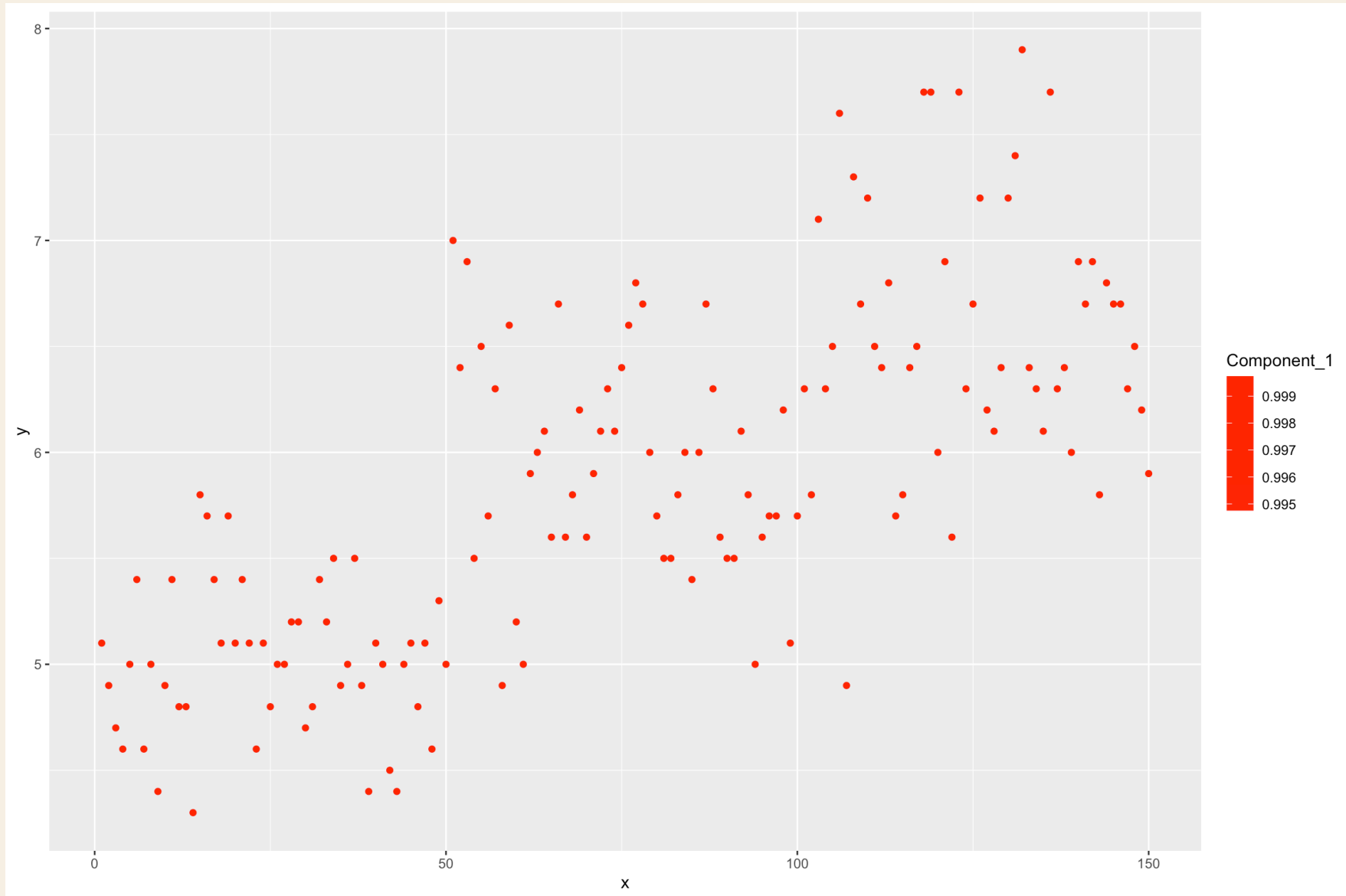
iris_density_plot = ggplot(iris_long, aes(x = Value)) +
  geom_histogram(
    aes(y = after_stat(density)),
    bins = 40,
    fill = "#e8edf0",
    color = "#374151",
    linewidth = 0.25
  ) +
  geom_line(
    data = iris_density,
    aes(x = Value, y = density, color = component),
    linewidth = 1.15,
    inherit.aes = FALSE
  ) +
  scale_color_manual(values = c("Component 1" = "#0b5cad", "Component 2" = "#b3212d")) +
  facet_wrap(~ Feature, scales = "free") +
  labs(x = "Value", y = "Density", color = NULL)

iris_density_plot
```

Iris Mixture Densities Plot



Iris Mixture Probabilities



Inspect Components

```
kable(sl_mixture$mc)  
kable(sw_mixture$mc)  
kable(pl_mixture$mc)  
kable(pw_mixture$mc)
```

```
sl_mixture$p_raw  
sw_mixture$p_raw  
pl_mixture$p_raw  
pw_mixture$p_raw
```

```
sl_mixture$p_density  
sw_mixture$p_density  
pl_mixture$p_density  
pw_mixture$p_density
```